

RECAP: 1) IID Process

$\{X_t, t: T\}$ - IID with common distribution $\Leftrightarrow F_{t_1, \dots, t_n}(x_1, \dots, x_n) = \prod F_{t_i}(x_i)$
- Every IID process is stationary

2) Bernoulli Process : A seq. of IID $\text{Ber}(p)$ RVs.

3) Arrival Process : 3 characterisations

- Arrival epochs : $\{S_n, n \in \mathbb{N}\}$

- Inter-arrival times : $\{X_n : n \in \mathbb{N}\}$

- Counting Process : $\{N_t : t \in \mathbb{R}^+\}$.

4) Renewal Process : Arrival Process with IID Inter-arrival times

Exponential Distribution and Memoryless Property

Consider a random variable X defined on $(\Omega, \mathcal{F}, \mathbb{P})$ s.t. $X \sim \exp(\lambda)$.

- PDF: $f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$
- CDF: $F_X(x) = 1 - e^{-\lambda x}, \quad x \in \mathbb{R}$
- Complementary CDF: $\mathbb{P}(X > x) = e^{-\lambda x}, \quad x \in \mathbb{R}$
- Memoryless Property: $\mathbb{P}(X > s+t | X > s) = \mathbb{P}(X > t), \quad s, t \in \mathbb{R}$
Prob. of waiting for an additional time of t given that the waiting time has already exceeded s is the same as the prob. of waiting for t from the beginning
$$\mathbb{P}(X > s+t) = \mathbb{P}(X > s) \mathbb{P}(X > t).$$

Qn: $X \sim \exp \Rightarrow X$ memoryless.

- For a continuous RV X , does the reverse implication hold?
i.e., is exponential distribution the only continuous distribution that has the memoryless property?

Qn: Are regular arrivals memoryless?

For example: a bus arrives at a bus stop A once every 15 minutes.

Qn: If the memoryless property is restricted to integer values i.e., if
 $\mathbb{P}(X > t+k | X > t) = \mathbb{P}(X > k)$ should hold for only for $k, t \in \mathbb{Z}^+$,
then which distribution has memoryless property?

Bernoulli process which has IID geometric inter-arrival times is like a discrete-time version of the Poisson process (which has IID exponential inter-arrival times)

Example: The $M/M/1$ queue

$\nwarrow$ service process
 $\uparrow$ arrival process
 $\searrow$ number of servers

M : memoryless
 D : Deterministic
 G : General

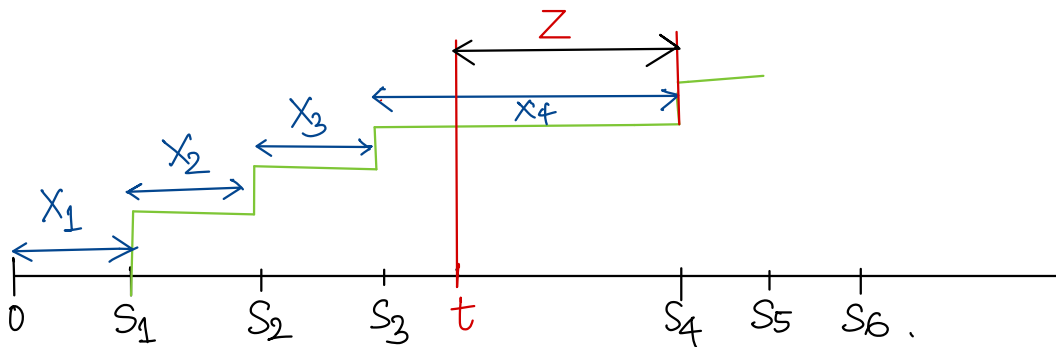
The first $M \Rightarrow$ arrival process is memoryless $\Rightarrow$ Poisson process (IID exp. distributed inter-arrival times).

The second $M \Rightarrow$ service process is memoryless $\Rightarrow$ IID exp. distributed service times.

1 $\rightarrow$ single server.

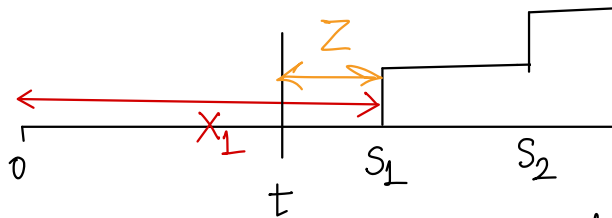
$M/M/1$ queue $\Rightarrow$ A queueing system with a Poisson arrival ($\exp(\lambda)$) and a single server that serves arriving customers in FCFS order with service times $\sim \exp(\mu)$. The service times are independent of each other and of the inter-arrival times.

Theorem: For a Poisson process of rate λ , and any given $t > 0$, the length of the interval from t until the first arrival after t is a positive RV Z with the CDF $F_Z(z) = 1 - \exp(-\lambda z)$, $z \geq 0$. This RV is independent of both N_t and the N_t arrival epochs $S_1, S_2, \dots, S_{N_t}$ before time t .



Implication: The portion of a Poisson process starting at an arbitrary time $t > 0$ is a probabilistic replica of the process starting at $t=0$; i.e., the time until the first arrival after t is a RV $\sim \exp(\lambda)$, and all subsequent inter-arrival times are independent of this first arrival epoch and of each other, and all these are distributed $\exp(\lambda)$.

Proof:



Let Z be the distance from t until the first arrival after t .

Let's first condition on $N_t = 0$

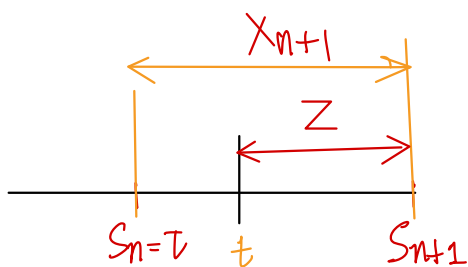
$$N_t = 0 \Rightarrow X_1 > t \text{ and } Z = X_1 - t$$

$$\begin{aligned} \Pr\{Z > z \mid N_t = 0\} &= \Pr\{X_1 > z + t \mid N_t = 0\} \\ &= \Pr\{X_1 > z + t \mid X_1 > t\} \\ &= \Pr\{X_1 > z\} = e^{-\lambda z} \end{aligned}$$

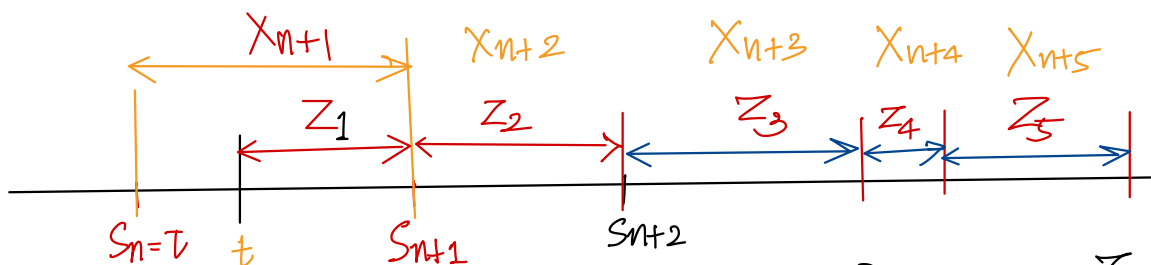
Now consider $N_t = n$ $\{n \geq 1\}$ and $S_n = \tau$ $\{\text{arbitrary } \tau \leq t\}$

$N_t = n$ and $S_n = \tau \leq t \Rightarrow$ the first arrival after t is the arrival at S_{n+1}

$$\begin{aligned} \therefore \Pr\{Z > z \mid N_t = n, S_n = \tau\} &\Rightarrow \Pr\{X_{n+1} > z + t - \tau \mid N_t = n, S_n = \tau\} \\ &= \Pr\{X_{n+1} > z + t - \tau \mid X_{n+1} > t - \tau, S_n = \tau\} \\ &= \Pr\{X_{n+1} > z + t - \tau \mid X_{n+1} > t - \tau\} \\ &= \Pr\{X_{n+1} > z\} = e^{-\lambda z} \end{aligned}$$



Sequence of inter-arrival times starting from t : $Z_1, Z_2, \dots$



Given $N_t = n$ and $S_n = \tau \leq t$, $Z_m = X_{m+n}$, for $m \geq 2$ $Z_1, Z_2, \dots$ IID exp RVs

Why is it called a Poisson process?

Ans: Counting Process N_t has a Poisson distribution.

$$f_{X_i}(x_i) = \begin{cases} \lambda e^{-\lambda x_i} & , x_i \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Inter-arrival times $X_1, X_2, X_3, \dots \stackrel{\text{iid}}{\sim} \exp(\lambda)$

Arrival epochs $S_1 = X_1, S_2 = X_1 + X_2, \dots, S_n = X_1 + X_2 + \dots + X_n$.

Joint pdf of $X_1, X_2, \dots, X_n$: $f_{X_1, \dots, X_n}(x_1, \dots, x_n) = \begin{cases} \lambda^n e^{-\lambda(x_1 + \dots + x_n)} & , (x_1, \dots, x_n) \in (\mathbb{R}^+)^n \\ 0 & , \text{otherwise} \end{cases}$

Find joint pdf of $S_1, \dots, S_n$

$$J = \left| \frac{\partial \underline{S}}{\partial \underline{X}} \right| = \begin{vmatrix} \frac{\partial S_1}{\partial X_1} & \frac{\partial S_1}{\partial X_2} & \dots & \frac{\partial S_1}{\partial X_n} \\ \frac{\partial S_2}{\partial X_1} & \frac{\partial S_2}{\partial X_2} & \dots & \frac{\partial S_2}{\partial X_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial S_n}{\partial X_1} & \frac{\partial S_n}{\partial X_2} & \dots & \frac{\partial S_n}{\partial X_n} \end{vmatrix} =$$

$$f_{S_1, \dots, S_n}(s_1, \dots, s_n) = \begin{cases} \lambda^n e^{-\lambda s_n} & , 0 < s_1 < s_2 < \dots < s_n \\ 0 & \text{otherwise} \end{cases}$$

The event $\{N_t = n\}$ is equivalent to the event $(S_1, \dots, S_n, S_{n+1}) \in A_{n,t}$, where

$$A_{n,t} = \left\{ (t_1, t_2, \dots, t_n, t_{n+1}) \in (\mathbb{R}^+)^{n+1} : 0 < t_1 < t_2 < \dots < t_n \leq t < t_{n+1} \right\}$$

The conditional pdf of $(S_1, \dots, S_{n+1})$ given that $\{N_t = n\}$ is obtained by starting with the joint pdf of $S_1, \dots, S_{n+1}$ on the set $\{t \in (\mathbb{R}^+)^{n+1} : 0 < t_1 < t_2 < \dots < t_{n+1}\}$ and setting it to zero outside $A_{n,t}$ and scaling it by $P(\{N_t = n\})$ on $A_{n,t}$

$$f_{S_1, \dots, S_{n+1}} | \{N_t = n\} (t_1, \dots, t_{n+1} | N_t = n) = \begin{cases} \frac{\lambda^{n+1} e^{-\lambda t_{n+1}}}{P(\{N_t = n\})} & , 0 < t_1 < \dots < t_n \leq t < t_{n+1} \\ 0 & , \text{otherwise} \end{cases}$$

We want $f_{S_1, \dots, S_n} | \{N_t = n\}$

Obtained by integrating $f_{S_1, \dots, S_{n+1}} | \{N_t = n\}$ over all values of S_{n+1} .

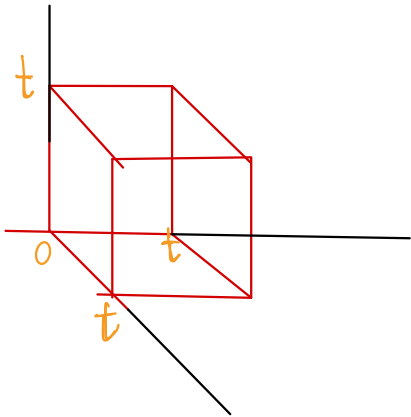
if $0 < t_1 < \dots < t_n \leq t$ does NOT hold, $f_{S_1, \dots, S_{n+1}} | \{N_t = n\} = 0$ $\forall$ values of S_{n+1} .

if $0 < t_1 < \dots < t_n \leq t$, then $f_{S_1, \dots, S_{n+1}} | \{N_t = n\} > 0$ for $S_{n+1} > t$

$\therefore$ For $0 < t_1 < \dots < t_n \leq t$

$$\begin{aligned} f_{S_1, \dots, S_n | N_t = n}(t_1, \dots, t_n | N_t = n) &= \int_t^\infty f_{S_1, \dots, S_{n+1}} | \{N_t = n\}(t_1, \dots, t_{n+1} | N_t = n) dt_{n+1} \\ &= \frac{1}{P(\{N_t = n\})} \int_t^\infty \lambda^{n+1} e^{-\lambda t_{n+1}} dt_{n+1} \\ &= \frac{1}{P(\{N_t = n\})} \lambda^n \int_t^\infty \lambda e^{-\lambda t_{n+1}} dt_{n+1} = \frac{\lambda^n e^{-\lambda t}}{P(\{N_t = n\})} \end{aligned}$$

$$\therefore f_{S_1, \dots, S_n | N_t = n}(t_1, \dots, t_n | N_t = n) = \begin{cases} \frac{\lambda^n e^{-\lambda t}}{P(\{N_t = n\})}, & 0 < t_1 < \dots < t_n \leq t \\ 0, & \text{otherwise} \end{cases}$$



The conditional density is constant on the set $\{(t_1, \dots, t_n) \in (\mathbb{R}^+)^n : 0 < t_1 < t_2 < \dots < t_n \leq t\}$ A_n^t

and the constant = $\frac{1}{\text{n-D volume of the set } A_n^t}$

$= \frac{t^n}{n!}$

$$\therefore \frac{\lambda^n e^{-\lambda t}}{P(\{N_t = n\})} = \frac{n!}{t^n} \Rightarrow P(\{N_t = n\}) = \frac{(\lambda t)^n e^{-\lambda t}}{n!}$$

ie $N_t \sim \text{Pos}(\lambda t)$